STPM 2019 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $g = 9.81$ m s^{-2}.

Section A [15 marks]

1. Which statement is **not** true about damped oscillations?

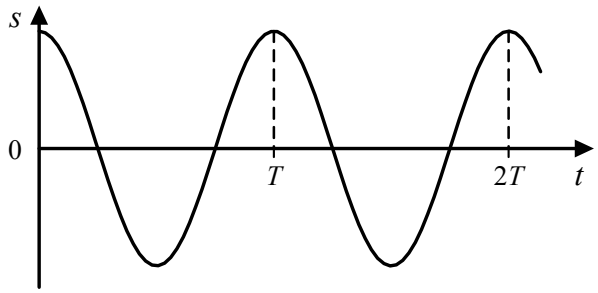
A In all damped oscillations, some of the mechanical energy is lost as heat.
B In an over damped oscillation, the object oscillates for a very long period.
C In an under damped oscillation, the object oscillates with decreasing amplitude.
D In a critically damped oscillatory system, the object moves to the equilibrium position in the shortest possible time and stops.

Answer: B

An over-damped system does not oscillate at all; it returns slowly to equilibrium. Statement B is not true.

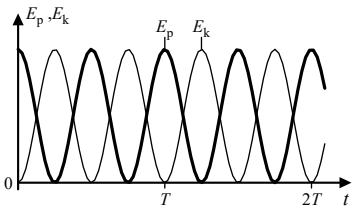
Explanation: Damping always converts mechanical energy into heat; critical damping returns to equilibrium fastest without oscillating.

2. A graph of displacement, s, against time, t, of a body performing simple harmonic motion with period, T, is shown in the diagram below.

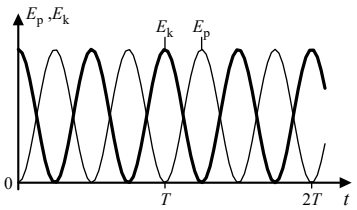


Which graph shows the variations of kinetic energy, E_k, and potential energy, E_P, of the body with time, t?

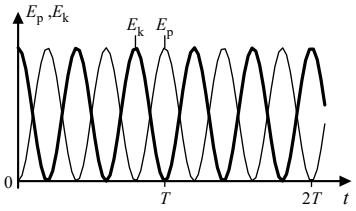
A



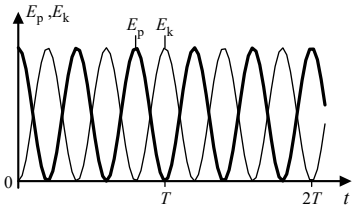
B



C



D



Answer: A

The displacement starts at its maximum, so at $t = 0$ the potential energy is maximum and the kinetic energy is zero.

Both energies repeat twice per period; at $t = T$ the body is again at maximum displacement, so E_p is maximum there (graph A).

Explanation: The two energies are always out of step: when one is maximum the other is zero, and their sum is constant.

3. The equation for a progressive wave is given by $y = 0.40 \sin (20t - 4.0x)$, where y and x are in metre and t is in second. The speed of the wave is

A $0.20\ \text{m s}^{-1}$ B $5.0\ \text{m s}^{-1}$ C $50\ \text{m s}^{-1}$ D $80\ \text{m s}^{-1}$

Answer: B

$$v = \frac{\omega}{k} = \frac{20}{4.0} = 5.0\ \text{m s}^{-1}$$

Explanation: Compare with $y = A \sin (\omega t - kx)$.

4. The intensity of sound wave at a distance of 3.0 m from a point source is $20\ \text{W m}^{-2}$. What is the intensity of the sound wave at a distance of 4.0 m from the source?

A $8.4\ \text{W m}^{-2}$ B $11\ \text{W m}^{-2}$ C $15\ \text{W m}^{-2}$ D $36\ \text{W m}^{-2}$

Answer: B

$$I_2 = I_1\left(\frac{r_1}{r_2}\right)^2 = 20\left(\frac{3.0}{4.0}\right)^2 = 11\ \text{W m}^{-2}$$

Explanation: A point source obeys the inverse square law.

5. A guitar string has length L and mass M. If the frequency of the first overtone of the string is f, what is the tension in the string?

A MLf^2 B $4MLf^2$ C ML^2f^2 D $4ML^2f^2$

Answer: A

First overtone: two loops, $\lambda = L$, so $v = fL$

$$v = \sqrt{\left(\frac{T}{\mu}\right)} \text{ with } \mu = \frac{M}{L} \Longrightarrow T = \mu v^2 = \frac{M}{L}(fL)^2 = MLf^2$$

Explanation: The first overtone is the second harmonic.

6. A police car sounds its siren of frequency 500 Hz and moves with constant speed of 40 m s^{-1} along a straight one-way road towards a speeding car. If the speeding car moves with a constant speed of 25 m s^{-1} away from the police car along the road, what is the frequency of the siren heard by the driver of the speeding car?
[The speed of sound in air is 320 m s^{-1}.]

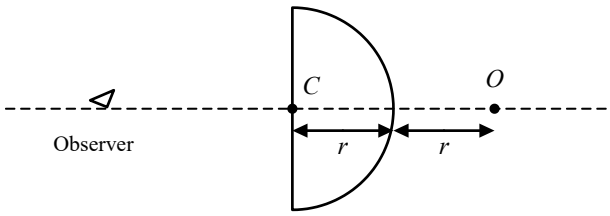
A 410 Hz B 480 Hz C 530 Hz D 620 Hz

Answer: C

$$f' = f\frac{v - v_o}{v - v_s} = 500\,\frac{320 - 25}{320 - 40} = 527 \text{ Hz} \approx 530 \text{ Hz}$$

Explanation: The source moves towards the observer (raises f) while the observer moves away (lowers f); the source is faster, so the net effect is an increase.

7. A hemispherical glass block of radius r with the centre of curvature C and refractive index 1.5 is shown in the diagram below.



An object, O, is placed at a distance r from the curved surface. The image formed by the curved surface is located at

A a distance of $2r$ to the right of O
B a distance of r to the right of O
C a distance of $2r$ to the right of C
D a distance of r to the right of C

Answer: A

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R} : \frac{1}{r} + \frac{1.5}{v} = \frac{0.5}{r}$$

$v = -3r$: a virtual image $3r$ from the curved surface, on the same side as O, i.e. $2r$ to the right of O

Explanation: Light from O enters the glass at the curved surface, which is convex towards O (radius $+r$). The object is too close for the surface to form a real image.

8. A thin plano-convex lens is made of glass of refractive index 1.5 and the curve surface has radius of curvature of 40 cm. What is the focal length of the lens?

A 8 cm B 16 cm C 40 cm D 80 cm

Answer: D

$$\frac{1}{f} = (1.5 - 1)\frac{1}{40} \Rightarrow f = 80 \text{ cm}$$

Explanation: The flat surface has an infinite radius of curvature.

9. Two sources emit waves which interfere in a region. Which condition is **not** necessary for a stable interference pattern to occur?

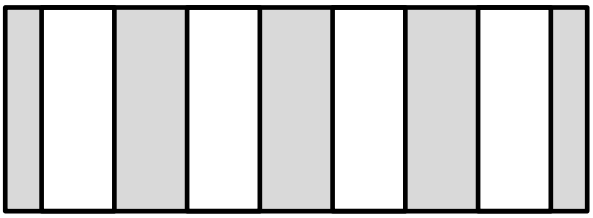
A The wavelengths of both waves must be equal.
B The frequencies of both waves must be equal.
C The amplitudes of both waves must be equal.
D The types of both waves must be similar.

Answer: C

The amplitudes need not be equal for a stable (fixed) pattern; equal amplitudes only make the dark fringes completely dark. Statement C is not necessary.

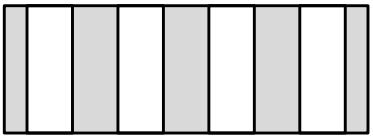
Explanation: The sources must be coherent: same frequency (and wavelength), constant phase difference and the same type of wave.

10. The interference fringes produced by a Young's two-slit experiment is shown in the diagram below.



If the distance between the two slits is increased, which diagram shows the correct interference fringes?

A



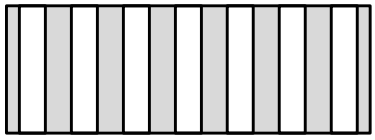
B



C



D



Answer: D

$$\text{Fringe separation } y = \frac{\lambda D}{d}\text{: increasing } d \text{ decreases } y$$

The fringes become narrower and closer together (D).

Explanation: The fringe width is inversely proportional to the slit separation.

11. A particle of mass m has kinetic energy E. What is the de Broglie wavelength of the particle?

A $h\sqrt{(2mE)}$ B $\sqrt{\frac{2mE}{h}}$ C $\frac{h}{\sqrt{(2mE)}}$ D $\sqrt{\frac{h}{2mE}}$

Answer: C

$$p = \sqrt{(2mE)} \Longrightarrow \lambda = \frac{h}{p} = \frac{h}{\sqrt{(2mE)}}$$

Explanation: Kinetic energy $E = p^2/2m$.

12. Which statement explains the existence of a minimum wavelength in the X-ray spectrum?

A An incident electron collides with the anode.
B An incident electron experiences deceleration during the collision.
C An incident electron knocks an electron out from the innermost shell.
D An incident electron radiates all its kinetic energy as a single photon during the collision.

Answer: D

The minimum wavelength corresponds to an electron converting all its kinetic energy into a single photon: $eV = hc/\lambda_{min}$.

Explanation: No photon can carry more energy than the electron had, so there is a sharp cut-off wavelength.

13. The atomic plane spacing for a crystal structure used in an X-ray diffractometer is 0.300 nm. If the wavelength of the incident X-ray is 0.154 nm, what is the maximum order of X-ray diffraction observed from the planes?

A 1 B 2 C 3 D 4

Answer: C

$$n \leq \frac{2d}{\lambda} = \frac{2(0.300)}{0.154} = 3.9 \Longrightarrow n_{max} = 3$$

Explanation: Bragg's law $2d \sin\theta = n\lambda$ with $\sin\theta \leq 1$.

14. Carbon-14 $^{14}_{6}C$ decays to Nitrogen-14 $^{14}_{7}N$ by emitting beta particle. What is the number of protons and electrons of nitrogen after the beta decay?

A 6 protons, 7 electrons
B 7 protons, 6 electrons
C 7 protons, 7 electrons
D 7 protons, 8 electrons

Answer: B

In β^- decay a neutron becomes a proton, so the nucleus has 7 protons. The atom still has the 6 electrons of the original carbon atom (the β particle is emitted).

Explanation: The nitrogen is formed as a positive ion, N^+, immediately after the decay.

15. The activity of 4.0 mole of radioisotope X is five times more than that of 1.0 mole of radioisotope Y. What is the ratio of the half-life of X to that of Y?

A $\frac{1}{20}$ B $\frac{4}{5}$ C 4 D 20

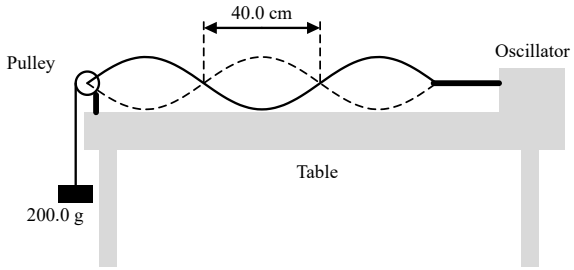
Answer: B

$$A = \lambda N \Longrightarrow \frac{\lambda_X}{\lambda_Y} = \frac{A_X/N_X}{A_Y/N_Y} = \frac{5/4}{1/1} = \frac{5}{4}$$
$$\frac{T_X}{T_Y} = \frac{\lambda_Y}{\lambda_X} = \frac{4}{5}$$

Explanation: Half-life is inversely proportional to the decay constant. The number of nuclei is proportional to the number of moles.

Section B [15 marks]

16. A string with one end attached to an oscillator of frequency 30.0 Hz and the other end to a mass of 200.0 g over a pulley to produce standing waves is shown in the diagram below.



(a) Describe the formation of standing waves in the string. [2]

Answer

The wave from the oscillator travels along the string and is reflected at the pulley — 1

The incident and reflected waves (same frequency, amplitude and speed, opposite directions) superpose to form a standing wave with fixed nodes and antinodes — 1

2

Explanation: The pulley end is a node.

(b) Determine the wavelength of the wave. [1]

Answer

Each loop is $\frac{\lambda}{2} = 40.0$ cm $\Longrightarrow \lambda = 0.800$ m — 1

1

Explanation: The 40.0 cm marked is the length of one loop (node to node).

(c) Calculate the speed of the wave in the string. [2]

Answer

$v = f\lambda = (30.0)(0.800)$ 1

$v = 24.0 \text{ m s}^{-1}$ 1

2

Explanation: The frequency of the standing wave is the oscillator frequency.

(d) Determine the mass per unit length of the string. [3]

Answer

$T = mg = (0.2000)(9.81) = 1.96 \text{ N}$ 1

$v = \sqrt{(\frac{T}{\mu})} \Longrightarrow \mu = \frac{T}{v^2} = \frac{1.96}{24.0^2}$ 1

$\mu = 3.41 \times 10^{-3} \text{ kg m}^{-1}$ 1

3

Total for question **8**

Explanation: The tension is the weight of the hanging mass.

17. A nuclear fusion reaction between lithium and deuterium is given by ${}^{6}_{3}\text{Li} + {}^{2}_{1}\text{H} \rightarrow 2X$.

[Atomic mass of ${}^{6}_{3}\text{Li}$ is 6.015123 u; atomic mass of deuterium is 2.014102 u; atomic mass of X is 4.002603 u; 1 u ≡ 931.5 MeV.]

(a) Determine X. [1]

Answer

${}^{6}_{3}\text{Li} + {}^{2}_{1}\text{H} \rightarrow 2\ {}^{4}_{2}\text{He}$: X is ${}^{4}_{2}\text{He}$ (an α particle) 1

1

Explanation: Nucleon number: $6 + 2 = 2 \times 4$; proton number: $3 + 1 = 2 \times 2$.

(b) Calculate the energy released. [4]

Answer

Mass before = 6.015123 + 2.014102 = 8.029225 u 1

Mass after = 2(4.002603) = 8.005206 u 1

$\Delta m = 0.024019$ u 1

$E = (0.024019)(931.5) = 22.4$ MeV 1

4

Explanation: Atomic masses can be used because the electron numbers balance on both sides.

(c) Explain why the reaction can only occur at very high temperature. [2]

Answer

Both nuclei are positively charged and repel each other strongly	1
They need very high kinetic energies (very high temperature) to overcome the Coulomb repulsion and come close enough for the strong nuclear force to act	1
	2
Total for question	**7**

Explanation: This is why fusion happens naturally only in the cores of stars.

Section C [30 marks]

18. (a) Define *simple harmonic motion*. [2]

Answer

The acceleration is directly proportional to the displacement from a fixed point (equilibrium position)	1
and always directed towards that point: $a = -\omega^2 x$	1
	2

Explanation: Both the proportionality and the direction are needed.

(b) A simple pendulum consists of a bob of mass 200 g and a string of length 1.50 m. The bob is displaced at an angle of 10.0° with the vertical and released at $t = 0$. The air resistance is negligible.

(i) Show that the bob moves in simple harmonic motion by considering the forces acting on it. [5]

Answer

Forces on the bob: weight mg and tension T; the restoring force along the arc is $mg \sin\theta$, towards the equilibrium position	1
$ma = -mg \sin\theta$	1
For small θ (10.0°), $\sin\theta \approx \theta = \frac{x}{l}$	1
$a = -\frac{g}{l}x$	1
g/l is constant, so $a \propto -x$: SHM with $\omega^2 = \frac{g}{l}$	1
	5

Explanation: The tension acts along the string and has no component along the arc.

(ii) The equation of motion for the bob can be represented by $x = A \sin(\omega t + \phi)$, where x is the displacement, A is the amplitude, ω is the angular frequency and ϕ is the phase angle. Calculate the values of A, ω and ϕ. [5]

Answer

$A = l\theta = (1.50)(10.0° \times \frac{\pi}{180})$ 1

$A = 0.262$ m 1

$\omega = \sqrt{(\frac{g}{l})} = \sqrt{(\frac{9.81}{1.50})} = 2.56$ rad s^{-1} 1

At $t = 0$, $x = A \Longrightarrow \sin \phi = 1$ 1

$\phi = \frac{\pi}{2}$ rad 1

5

Explanation: The bob is released from rest at maximum displacement, so $x = A \cos \omega t = A \sin(\omega t + \pi/2)$.

(iii) Determine the speed of the bob when the displacement is 10.0 cm. [2]

Answer

$v = \omega\sqrt{(A^2 - x^2)} = 2.56\sqrt{(0.262^2 - 0.100^2)}$ 1

$v = 0.619$ m s^{-1} 1

2

Explanation: Use the displacement in metres (10.0 cm = 0.100 m).

(iv) State a difference in the motion of the pendulum if it were to be brought to the Moon. [1]

Answer

The period is longer (lower frequency) because g on the Moon is smaller ($T = 2\pi\sqrt{(\frac{l}{g})}$) 1

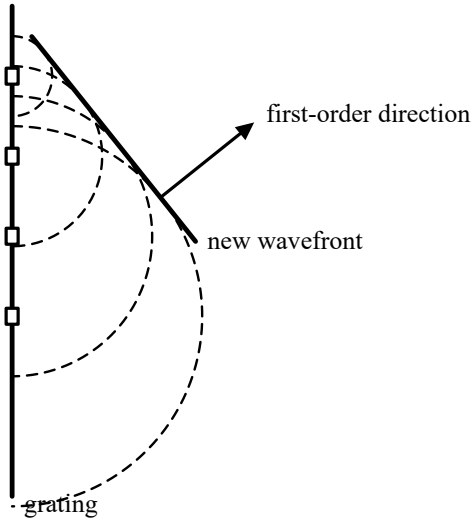
1

Total for question **15**

Explanation: The period of a simple pendulum does not depend on the mass of the bob.

19. (a) Explain, with the aid of a diagram, the production of bright fringes in a diffraction grating by using Huygens's principle. [4]

Answer



Each slit acts as a source of secondary wavelets (Huygens's principle)	1
In certain directions the wavelets from adjacent slits have a path difference $d \sin \theta = n\lambda$	1
so they arrive in phase and interfere constructively	1
The common tangent to these wavelets forms a new plane wavefront travelling in that direction, giving a bright fringe (order n)	1
	4

Explanation: In the diagram the wavelets from neighbouring slits differ in radius by one wavelength, so their common tangent is the first-order wavefront.

(b) State **two** differences between two-slit interference pattern and single-slit diffraction pattern. [2]

Answer

Two-slit pattern: bright fringes equally spaced and of (nearly) equal width; single slit: central bright fringe twice as wide as the others	1
Two-slit fringes have (nearly) equal intensity; in single-slit diffraction the central maximum is much brighter and the side maxima fall off rapidly	1
	2

Explanation: Both patterns come from superposition of waves; they differ in the number of sources.

(c) A monochromatic light of frequency 6.00×10^{14} Hz is used in a Young's two-slit experiment. The fringe separation is 0.75 mm and the distance between the screen and the slits is 1.80 m.

(i) Calculate the distance between the two slits. [3]

Answer

$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{6.00 \times 10^{14}} = 5.00 \times 10^{-7}$ m — 1

$d = \frac{\lambda D}{x} = \frac{(5.00 \times 10^{-7})(1.80)}{0.75 \times 10^{-3}}$ — 1

$d = 1.20 \times 10^{-3}$ m — 1

3

Explanation: Find the wavelength from the frequency first.

(ii) If a thin film of glass of refractive index 1.50 is placed in front of one of the slits, the original central bright fringe will become the 10th bright fringe. Determine the thickness of the thin film. [3]

Answer

The film adds an extra optical path $(n - 1)t$ — 1

The central fringe moves by 10 fringes: $(n - 1)t = 10\lambda$ — 1

$t = \frac{10(5.00 \times 10^{-7})}{0.50} = 1.00 \times 10^{-5}$ m — 1

3

Explanation: Each fringe shift corresponds to an extra optical path of one wavelength.

(iii) If one of the slits is covered, a single slit diffraction pattern is observed and the width of the central bright fringe is 1.0 cm. Determine the width of the single slit. [3]

Answer

Width of central maximum $= \frac{2\lambda D}{a}$ — 1

$a = \frac{2(5.00 \times 10^{-7})(1.80)}{1.0 \times 10^{-2}}$ — 1

$a = 1.80 \times 10^{-4}$ m — 1

3

Total for question **15**

Explanation: The first minima are at $\theta = \pm\lambda/a$, so the central maximum is $2\lambda D/a$ wide.

20. (a) State the Bohr's postulate which relates to quantisation principle. [1]

Answer

The angular momentum of the electron is quantised: $mvr = \frac{nh}{2\pi}$, $n = 1, 2, 3, \ldots$ — 1

1

Explanation: Electrons can only occupy orbits that satisfy this condition.

(b)

(i) Derive the expression for the radius of the first orbit of hydrogen atom. [5]

Answer

Coulomb force provides the centripetal force: $\frac{mv^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2}$ — 1

Quantisation: $mvr = \frac{nh}{2\pi} \Longrightarrow v = \frac{nh}{2\pi mr}$ — 1

Substitute: $\frac{n^2h^2}{4\pi^2mr^2} = \frac{e^2}{4\pi\varepsilon_0 r}$ — 1

$r = \frac{\varepsilon_0 n^2 h^2}{\pi m e^2}$ — 1

First orbit $n = 1$: $r_1 = \frac{\varepsilon_0 h^2}{\pi m e^2}$ — 1

5

Explanation: This is the Bohr radius.

(ii) Determine the radius of the first orbit of hydrogen atom. [1]

Answer

$r_1 = \frac{(8.85 \times 10^{-12})(6.63 \times 10^{-34})^2}{\pi(9.11 \times 10^{-31})(1.60 \times 10^{-19})^2} = 5.31 \times 10^{-11}$ m — 1

1

Explanation: About 0.053 nm.

(iii) Determine the kinetic energy and potential energy of the electron in the first orbit of hydrogen atom. [4]

Answer

$K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\varepsilon_0 r}$ 1

$K = \frac{(1.60 \times 10^{-19})^2}{8\pi(8.85 \times 10^{-12})(5.31 \times 10^{-11})} = 2.17 \times 10^{-18}$ J 1

$U = -\frac{e^2}{4\pi\varepsilon_0 r} = -2K$ 1

$U = -4.34 \times 10^{-18}$ J 1

4

Explanation: The total energy $K + U = -2.17 \times 10^{-18}$ J $= -13.6$ eV, the ground-state energy of hydrogen.

(c) Describe the formation of emission line spectra of a stable hydrogen atom. [4]

Answer

Atoms are excited (by heating or electrical discharge) so that electrons move to higher energy levels 1

The excited states are unstable; electrons fall back to lower levels 1

Each transition emits a photon of energy $hf = E_i - E_f$ 1

Only discrete energy levels exist, so only certain wavelengths are emitted: a line spectrum (e.g. Lyman, Balmer series) 1

4

Total for question **15**

Explanation: Each line in the spectrum corresponds to one particular pair of energy levels.